A Bandit Approach for Quantum Multi-Path Selection with Stochastic Blocking and Rate-Fidelity Tradeoff

***Abstract*—E2E entanglement distribution in quantum networks depends critically on the selection of paths between source-destination pairs. Path selection is therefore important since it directly determines the throughput, resource efficiency, and reliability of E2E entanglement distribution. Previous works often assume that the statistics of rates, costs, and fidelities are known a priori, and neglect the joint impact of rate-fidelity tradeoff and stochastic path blocking. However, in practical quantum networks, these statistics are typically unknown and must be learned online, and ignoring the coupled effects of rate-fidelity tradeoff and stochastic blocking leads to suboptimal or infeasible decisions. In this paper, we consider an online multi-path selection problem in quantum networks that maximizes the long-term average weighted rate subject to a per-slot cost budget and a fidelity threshold, while accounting for the inherent rate-fidelity tradeoff and stochastic path blocking. We propose OPS-UCB, an online learning algorithm that maintains confidence bounds for rate, cost, and fidelity and invokes an approximation oracle to select near-optimal feasible path sets, and prove that it achieves sublinear regret and sublinear long-term violation. Simulations on a 50-path quantum network demonstrate that OPS-UCB achieves near-zero long-term violation with competitive reward performance against an ideal benchmark.**

***Index Terms*—Quantum networks, multi-armed bandit, online path selection, rate-fidelity tradeoff, stochastic blocking**

I. Introduction

In quantum networks, quantum path selection between source-destination (SD) pairs plays a central role in determining the ebit rate of entanglement distribution, the efficiency of scarce quantum resource usage, and the fidelity of the delivered entanglement, so as to facilitate applications such as distributed quantum computing [1] and quantum sensing [2], etc. In practice, quantum path selection is impacted by several factors: probabilistic entanglement generation, expensive quantum memories, and the fidelity of a quantum path must be above a certain threshold.

Existing works on quantum network path selection, however, exhibit several limitations. First, many approaches assume that the statistics of ebit rates, costs, and fidelities are known a priori and focus on offline optimization under full information [3], [4], whereas in reality these quantities are unknown, depending on time-varying channel conditions, resource contention, and hardware noise. Second, while learning-based approaches address this unknown-statistics challenge [5]–[7], they typically do not enforce budget and fidelity constraints. Third, existing works often neglect the intrinsic *stochastic blocking* of quantum paths: after a path is used for entanglement distribution, its quantum resources are consumed and require a non-negligible amount of time to be re-established, during which the path becomes unavailable for future requests. This recovery time is random and governed by the success probabilities of entanglement generation and swapping, which determines the number of attempts required before the path is re-established [8], [9].

To address the limitations above, we consider an online path selection problem that accounts for the stochastic path blocking. The objective is learn the path parameters and select multiple paths for multiple SD pairs to maximize the total weighted rate, subject to a cost budget and a fidelity threshold. The cost budget reflects limited quantum resources such as entanglement consumption or quantum memory occupation; exceeding it risks resource exhaustion or network congestion. The fidelity threshold is required because a quantum state with fidelity below the threshold becomes unusable for applications like entanglement distillation [10] or quantum key distribution [11].

To solve this problem, we propose an online learning algorithm, Online Quantum Multi-Path Selection (OPS-UCB). The algorithm first observes the set of available (unblocked) paths. For each path, it maintains empirical averages of ebit rate, cost, and fidelity, and constructs optimistic estimates for rate and fidelity, and a pessimistic estimate for cost. The algorithm then invokes an approximation oracle over the available paths: the oracle returns a subset of paths whose estimated cost and fidelity meets the constraints, while approximately maximizing the estimated ebit rate. After the paths have been utilized, the observed rate, cost, and fidelity are used to update the corresponding statistics. Each selected path then becomes blocked for a random duration to be re-established and recovered, during which it remains unavailable. We prove that OPS-UCB achieves sublinear regret relative to an optimal dynamic policy, and sublinear total long-term violation that combines both cost overturns and fidelity shortfalls. Simulations demonstrate that OPS-UCB achieves competitive reward performance against an ideal benchmark with near-zero total long-term violation. Compared with baselines, OPS-UCB attains the highest reward among all constraint-aware algorithms while keeping violation growth sublinear. Sensitivity analysis under varying cost budgets, fidelity thresholds, and delays further confirms the robustness of the proposed approach.

The main contributions of this paper are:

We formulate an online quantum multi-path selection problem to maximize weighted rate, subject to a per-slot cost budget and

a fidelity threshold and propose OPS-UCB, an online learning algorithm that accommodates stochastic blocking and selects feasible subsets for multiple SD pairs.

We prove that OPS-UCB achieves sublinear regret and sublinear total long-term violation, where violation aggregates both cost overturns and fidelity shortfalls.

We demonstrate using simulations on a 50-path quantum network that OPS-UCB achieves near-zero long-term violation with competitive reward against ideal benchmark, and its robustness under varying parameters.

The remainder of this paper is organized as follows. Sec. II reviews related work. Sec. III formulates the problem. Sec. IV presents the online algorithm and the theoretical analysis. Sec. V reports simulation results. Sec. VI concludes the paper and discusses future work.

II. Related Work

Quantum network path selection has been studied from several perspectives. A primary line focuses on ebit rate or throughput maximization. Early approaches assume known statistics and adapt classical shortest-path algorithms to quantum repeater networks [3]. More recent works address this objective under fidelity constraints: fidelity-based path selection for sum-rate maximization under a fidelity threshold [12], redundant entanglement provisioning and selection for throughput maximization [13], and online path selection under fidelity thresholds via mean-variance rewards [14]. A parallel line studies resource cost and budget constraints. Adaptive user-centric path selection maximizes entanglement success rate under a long-term budget via per-slot Lyapunov optimization [15], while bounded-cost successive elimination addresses cost-aware link selection [16]. Finally, temporal path unavailability has been characterized by blocking probability in mixed-efficiency repeater networks [17], and protection-path based resource allocation has been proposed to reduce blocking in QKD networks [18]. These works, however, typically assume either known statistics or single-shot offline optimization.

A commonly used framework for path selection is bandit learning. Introduced by Chen et al. [19], combinatorial multi-armed bandit (CMAB) provides a principled foundation for sequential decision-making under combinatorial constraints, and has been extended to semi-bandit feedback [20], general reward functions [21], and approximation oracles for NP-hard offline problems [22]. Classical CMAB, however, focuses on per-round reward maximization without enforcing constraints that accumulate over time. Liu et al. [23] address this by introducing combinatorial bandits with linear long-term constraints, and prove that UCB-LP achieves logarithmic regret and zero expected violation. Aggarwal et al. [24] propose a bi-criteria framework that converts offline approximation algorithms into online CMAB algorithms with sublinear regret and cumulative constraint violation guarantees. Different from previous works on bandit learning, we account for stochastic blocking of arms caused by resource recovery.

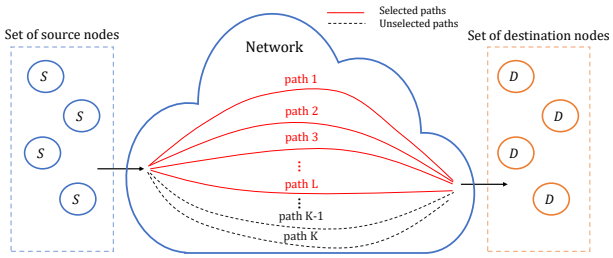


Fig. 1: Quantum network model for online path selection.

In bandit learning, another related but distinct models include sleeping bandits [27], where arms may be unavailable in certain rounds due to exogenous factors, and mortal bandits [28], where arms have finite lifetimes. Closest to our work is the blocking bandits model, where arms become unavailable for a random duration after each pull. Basu et al. [25] introduced deterministic blocking; Atsidakou et al. [26] extended it to combinatorial actions with stochastic delays, proving greedy approximation ratios and regret bounds. Although blocking bandits considers recovery delays, they neglect long-term cost of arms and application-specific constraint.

III. Problem Formulation

As shown in Fig. 1, consider a quantum network with a set of source nodes and a set of destination nodes. There are K precomputed end-to-end paths connecting the source and destination nodes. Let $[K] = \{1, 2, .., K\}$ denote the set of all paths. We consider a discrete-time model where time is partitioned into slots indexed by $t = 1, 2, ..., T$, where T is the time horizon. Each time slot corresponds to one round of path selection and quantum operation. Let $[T] = \{1, 2, ..., T\}$ denote the set of all time slots.

At each time slot t, the network needs to select L paths from the K paths ($L < K$) to serve L concurrent connection requests from the K SD pairs. Each selected path is assigned to exactly one request. The set of selected paths at time slot t is denoted by $\mathcal{P}_t \subset \{1, 2, ..., K\}$ with $|\mathcal{P}_t| = L$. Each path i is associated with a positive weight w_i reflecting its priority, which can represent the importance of the SD pair it serves or the service level of the traffic it carries; paths with higher weights are considered more important. For each selected path $i \in \mathcal{P}_t$, the observed path parameters, the ebit rate $r_i(t) \in [0, 1]$, usage cost $\mu_i(t) \in [0, 1]$, and end-to-end fidelity $F_i(t) \in [0, 1]$ at time slot t, are modeled as random variables drawn from fixed but unknown distributions. Denote their expectations as $r_i = \mathbb{E}[r_i(t)]$, $\mu_i = \mathbb{E}[\mu_i(t)]$, and $F_i = \mathbb{E}[F_i(t)]$. We assume that the rate, cost, and fidelity of a path are independent of each other. For convenience, we define the vectors of these quantities over all paths as $\boldsymbol{r}(t) = (r_1(t), ..., r_K(t))$, $\boldsymbol{\mu}(t) = (\mu_1(t), ..., \mu_K(t))$ and $\boldsymbol{F}(t) = (F_1(t), ..., F_K(t))$.

Whenever path i is selected at time slot t, it becomes blocked (unavailable) for $D_i(t) \in \mathbb{N}_{\geq 0}$ subsequent slots, namely, it cannot be selected again within the interval $[t, ..., t+$

$D_i(t)$). We model the blocking time $D_i(t)$ as a random variable drawn from a fixed but unknown path-dependent distribution with mean $d_i = \mathbb{E}[D_i(t)]$ and bounded by $[0, D_{\max}]$. A path with $D_i(t) = 0$ experiences no blocking and $D_{\max}$ denotes the maximum delay.

In the following, we formulate the online quantum multi-path selection problem.

Objective. Since each path i is assigned a weight w_i reflecting its priority, the total weighted ebit rate at time slot t is $\sum_{i\in\mathcal{P}_t} w_i r_i(t)$. Our objective is find paths to maximize the long-term average total weighted rate over T time slots.

$$\max_{\mathcal{P}_t, t\in[T]} \frac{1}{T}\sum_{t=1}^{T}\sum_{i\in\mathcal{P}_t} w_i r_i(t). \tag{1}$$

Availability constraint. Let $\mathcal{A}_t = \{i \in [K] \mid \forall s < t, \mathbb{I}(i \in \mathcal{P}(s)) \cdot \mathbb{I}(D_i(s) \geq t - s) = 0\}$ denote the set of available (unblocked) paths at the beginning of time slot t, where $\mathbb{I}(\cdot)$ is the indicator function that equals 1 when its argument is true and 0 otherwise. The paths must be selected from the available (unblocked) paths. That is,

$$\mathcal{P}_t \subseteq \mathcal{A}_t,\ \forall t \in [T]. \tag{2}$$

Budget constraint. Since each path is associated with a cost, the budget constraint is to ensure that the total cost incurred by the selected paths must not exceed a given budget B at every time slot:

$$\sum_{i\in\mathcal{P}_t} \mu_i(t) \leq B,\ \forall t \in [T]. \tag{3}$$

Fidelity constraint. To ensure the quality of the quantum communication, the end-to-end fidelity of each selected path must meet the application requirement. Therefore, we enforce that at every time slot, every chosen path $i \in \mathcal{P}_t$ must satisfy the fidelity threshold:

$$F_i(t) \geq F_{\min},\ \forall i \in \mathcal{P}_t, t \in [T], \tag{4}$$

where threshold $F_{\min}$ denotes the minimum fidelity requirement and $F_{\min} \geq 0.5$. This is because a fidelity below 0.5 cannot certify the presence of entanglement [29].

To establish a theoretical performance benchmark, we consider an optimal dynamic policy Π^* that knows the distributions of $r_i(t)$, $\mu_i(t)$, $F_i(t)$ and $D_i(t)$. This policy is dynamic because its decisions must adapt to the stochastically evolving set of available paths, which varies over time due to prior path selections and random blocking durations. Let $\mathcal{P}_t^*$ represent the set of paths selected by policy Π^* at time slot t. Let z_i denote the expected selecting rate of path i under the optimal dynamic policy Π^* and z_i can be expressed as

$$z_i = \mathbb{E}\left[\frac{1}{T}\sum_{t=1}^{T}\mathbb{I}(i \in \mathcal{P}_t^*)\right]. \tag{5}$$

Then the optimal expected cumulative reward that can be achieved by Π^* is:

$$\mathrm{Rew}^*(T) = T\sum_{i=1}^{K} w_i r_i z_i. \tag{6}$$

At each time slot t, an online algorithm selects a set $\mathcal{P}_t$ of L paths based solely on the partial observations from its previous selections. The algorithm does not know the underlying distributions of the path parameters, and more importantly, it can only observe the realized rate, cost, and fidelity for the paths that are actually selected; no observations are received for the unselected paths. Thus, the algorithm must learn them while selecting the paths.

In the full-information setting, where the true distributions of rate, cost, fidelity, and blocking delays are known and only the stochastic realizations and the evolving availability set remain uncertain, the quantum multi-path selection defined by (1)–(4) is a combinatorial optimization problem. It is well-established this problem is NP-hard and finding the exact solution is intractable. Hence, instead of assuming an exact solver, we introduce the following (α, β)-approximation oracle, which returns a feasible solution that is near-optimal with high probability:

Definition 1 ((α, β)-approximation oracle)**.** At time slot t, given a set of available paths $\mathcal{A}_t$ with $|\mathcal{A}_t| > L$, a reward vector $\boldsymbol{r}$, a cost vector $\boldsymbol{\mu}$, and a fidelity vector $\boldsymbol{F}$, denote the super set $\mathfrak{P}(\mathcal{A}_t) = \{\mathcal{P} \subseteq \mathcal{A}_t : |\mathcal{P}| = L, \sum_{i\in\mathcal{P}} \mu_i \leq B, F_i \geq F_{\min}, \forall i \in \mathcal{P}\}$. $f^{\alpha,\beta}(\mathcal{A}_t, \boldsymbol{r}, \boldsymbol{\mu}, \boldsymbol{F})$ is an (α, β)-approximation oracle that yields a set of paths $\mathcal{Q} \in \mathfrak{P}(\mathcal{A}_t)$ such that

$$\Pr\left[\sum_{i\in\mathcal{Q}} w_i r_i \geq \alpha \cdot \max_{\mathcal{P}\in\mathfrak{P}(\mathcal{A}_t)} \sum_{i\in\mathcal{P}} w_i r\right] \geq \beta. \tag{7}$$

A representative instance of such an (α, β)-approximation oracle is FPTAS [30], whose parameters satisfy $\alpha = 1 - \epsilon$ and $\beta = 1$, where $\epsilon > 0$ is the parameter that controls the tradeoff between approximation accuracy and computational cost. We consider a feasible approximately optimal algorithm, $\widetilde{\mathrm{OPT}}$, that repeatedly invokes the (α, β)-approximation oracle over the available paths with true averages of path parameters to select the path subsets. As shown in [26], this algorithm can achieve an asymptotic approximation ratio of $\frac{\alpha\beta}{1+\alpha\beta}$ with respect to the optimal dynamic policy Π^*, i.e., the expected cumulative reward of $\widetilde{\mathrm{OPT}}$ is $\frac{\alpha\beta}{1+\alpha\beta}\mathrm{Rew}^*(T)$.

In our bandit setting, we adopt this ratio as a benchmark discount factor and define the regret accordingly:

$$\mathrm{Reg}(T) = \left[\frac{\alpha\beta}{1+\alpha\beta}\mathrm{Rew}^*(T) - \mathbb{E}\Big[\sum_{t=1}^{T}\sum_{i\in\mathcal{P}_t} w_i r_i(t)\Big]\right]^+, \tag{8}$$

where $[\cdot]^+ = \max(\cdot, 0)$ is the positive-part operator, which ensures that only the shortfall relative to the discounted benchmark is counted; any episode in which the algorithm outperforms the benchmark contributes zero rather than negative regret. The first term is the discounted optimal expected reward, which serves as the performance benchmark; the second term is the expected cumulative reward actually obtained by the online algorithm.

To quantify how well the algorithm respects both constraints, we define the total long-term violation as the sum of the cost violation and the fidelity violation.

$$V(T) = \mathbb{E}\left[\left(\sum_{t=1}^{T}(\sum_{i\in\mathcal{P}_t}\mu_i(t) - B)\right)^+\right] + \mathbb{E}\left[\left(\sum_{t=1}^{T}\sum_{i\in\mathcal{P}_t}[F_{\min} - F_i(t)]\right)^+\right], \tag{9}$$

where both the cost and the fidelity term apply the positive-part operator to the horizon-wide cumulative sum to ensure that violation is never negative. We adopt this long-term definition rather than a per-round guarantee since in our unknown-distribution setting the learner cannot identify a feasible action for every slot before it has gathered sufficient samples; per-round feasibility would require prior knowledge of the true distributions that we do not assume.

The objective is to design an online algorithm that selects paths $\mathcal{P}_t$ for $t \in [T]$ and achieves sublinear regret, i.e., $\mathrm{Reg}_\rho(T) = o(T)$, while ensuring that the total violation is also sublinear, i.e., $V(T) = o(T)$. This implies that the average performance of the algorithm converges to the ρ-approximate benchmark, and both the average cost exceeds and the average fidelity deficit vanish as T grows.

IV. Algorithm and Analysis

Algorithm 1 outlines the detailed procedures of OPS-UCB. Specifically, the algorithm begins with an initialization phase, where for each path i, the counter $n_i(t)$, representing the number of times path i has been selected up to time slot t, is initialized to $n_i(0) = 1$ after selecting each path once to obtain initial empirical averages of the rate, cost and fidelity are set accordingly (Lines 1-4). At time slot t, after observing the set of currently available (unblocked) paths $\mathcal{A}_t$ (Line 6), the algorithm computes a confidence radius $\mathrm{rad}_i(t)$ for each available path i, which depends on its selection count up to the previous slot (Line 8). With this radius, it constructs optimistic Upper Confidence Bound (UCB) estimates, $\hat{r}_i(t)$ and $\hat{F}_i(t)$ for the rate and fidelity respectively, alongside a pessimistic Lower Confidence Bound (LCB) estimate, $\check{\mu}_i(t)$, for the cost (Lines 9-11). This encourages exploration of paths with potentially high rate, high fidelity, and low cost. Subsequently, the algorithm invokes the (α, β)-approximation oracle over the available set $\mathcal{A}_t$, using these constructed estimates; the oracle then returns a subset $\mathcal{P}_t$ of exactly L paths that satisfies the estimated cost and fidelity constraints while approximately maximizing the estimated weighted rate (Line 13). Finally, after the paths are selected, the true values of rate, cost, and fidelity are observed for each chosen path; the algorithm updates their respective empirical averages and increments the selection counters (Lines 14-20).

Based on this confidence-based exploration-exploitation mechanism, we now establish theoretical guarantees on both the regret bound and the constraint-satisfaction performance of our algorithm.

Algorithm 1 Online Quantum Multi-Path Selection (OPS-UCB)

Require: K, L, T, $f^{\alpha,\beta}$
Ensure: Set of selected paths $\mathcal{P}_t$ for each time slot t
1: **for** $i \in \{1, \ldots, K\}$ **do**
2: Select path i and observe initial information at $t = 0$
3: $n_i(0) \leftarrow 1$, $\bar{r}_i(0) \leftarrow r_i(0)$, $\bar{\mu}_i(0) \leftarrow \mu_i(0)$, $\bar{F}_i(0) \leftarrow F_i(0)$
4: **end for**
5: **for** $t = 1$ to T **do**
6: Obtain the set of available (unblocked) paths $\mathcal{A}_t$
7: **for** $i \in \mathcal{A}_t$ **do**
8: $\mathrm{rad}_i(t) \leftarrow \sqrt{\frac{3\ln t}{2n_i(t-1)}}$ ▷ Confidence radius
9: $\hat{r}_i(t) \leftarrow \bar{r}_i(t-1) + \mathrm{rad}_i(t)$ ▷ Rate estimate
10: $\check{\mu}_i(t) \leftarrow \bar{\mu}_i(t-1) - \mathrm{rad}_i(t)$ ▷ Cost estimate
11: $\hat{F}_i(t) \leftarrow \bar{F}_i(t-1) + \mathrm{rad}_i(t)$ ▷ Fidelity estimate
12: **end for**
▷ Call the (α, β)-approximation oracle in Definition 1
13: $\mathcal{P}_t \leftarrow f^{\alpha,\beta}(\mathcal{A}_t, \hat{\boldsymbol{r}}(t), \check{\boldsymbol{\mu}}(t), \hat{\boldsymbol{F}}(t))$
14: **for** $i \in \mathcal{P}_t$ **do**
15: Select path i for a connection request and observe $r_i(t)$, $\mu_i(t)$, $F_i(t)$
16: $n_i(t) \leftarrow n_i(t-1) + 1$
17: $\bar{r}_i(t) \leftarrow \frac{\bar{r}_i(t-1)n_i(t-1) + r_i(t)}{n_i(t)}$
18: $\bar{\mu}_i(t) \leftarrow \frac{\bar{\mu}_i(t-1)n_i(t-1) + \mu_i(t)}{n_i(t)}$
19: $\bar{F}_i(t) \leftarrow \frac{\bar{F}_i(t-1)n_i(t-1) + F_i(t)}{n_i(t)}$
20: **end for**
21: **end for**

We first state a regret bound that characterizes the algorithm's learning performance relative to an optimal dynamic policy that knows the underlying distributions. We then provide a constraint violation bound that quantifies the algorithm's ability to respect the per-slot cost budget and fidelity threshold over the long run. Both bounds are sublinear in the time horizon T, implying that the average performance converges to the benchmark and the average constraint violation vanishes as T grows. Proof sketches are provided below.

Theorem 1. For the online algorithm with an (α, β)-approximation oracle, the ρ-approximate regret, with $\rho = \frac{\alpha\beta}{1+\alpha\beta}$, satisfies the following distribution-independent bound:

$$\mathrm{Reg}_\rho(T) \leq \frac{14\sqrt{KLT\ln T}}{1+\alpha\beta} + \frac{K}{1+\alpha\beta}(2 + \pi^2\Delta_{\max}) + \mathcal{O}(D_{\max}\cdot K).$$

where $\Delta_{\max}$ denotes the maximum possible suboptimality gap described in the Appendix.

Proof Sketch. The proof proceeds in three conceptual steps. First, the ρ-approximate regret is reduced to the cumulative instantaneous regret—the per-slot loss incurred by selecting a

suboptimal set due to estimation uncertainty—plus a blocking overhead that accounts for the temporary unavailability of selected paths. Second, the instantaneous regret is decomposed according to two sources of error: the oracle may fail to return an α-approximate solution, and the confidence bounds may fail to contain the true means. The former contributes at most $(1-\beta)$ per slot to the expected loss, while the latter is controlled by the probability that the Hoeffding-based confidence intervals do not hold [31]. Summing these failure probabilities over T slots yields the $\pi^2 K\Delta_{\max}$ term. Third, on the good event that both the oracle succeeds and all confidence bounds hold, any selected set that is suboptimal must have its suboptimality gap bounded by the sum of the confidence radii of its constituent paths. A sampling threshold argument shows that the cumulative contribution of such gaps over time is at most $\mathcal{O}(\sqrt{KLT\ln T})$. Combining the three parts yields the stated bound. The complete proof is provided in the Appendix. □

Theorem 2. Under the same conditions as **Theorem 1**, the total long-term violation $V(T)$ defined in (9) satisfies the following sublinear bound:

$$V(T) \le 2\sqrt{24KLT\ln T} + 2\pi^2 KL + 2L\sqrt{T}.$$

Proof Sketch. At each time slot, the oracle guarantees that the estimated cost of the selected set does not exceed the budget, and the estimated fidelity of every selected path is above the threshold. Therefore, the actual violations are bounded by the deviations between the true values and their estimates. We decompose these deviations into three components. The first is the stochastic fluctuation of the realized values around their means; this forms a martingale difference sequence with bounded increments, so its positive part accumulates at rate $L\sqrt{T}$ by the Azuma-Hoeffding inequality. The second is the error between the empirical means and the true means, which is bounded with high probability by the confidence radii as established by the Hoeffding concentration inequality [31]. The third is the sum of the confidence radii themselves over all selected paths; by a standard counting argument, this accumulates to $\mathcal{O}(\sqrt{KLT\ln T})$. Summing these three bounds for both cost and fidelity violations, and taking the positive parts, yields the stated sublinear total violation bound. The complete proof is provided in the Appendix. □

V. Experiments

A. Experiment Setup

1) Parameter Setting: We simulate a quantum network with $K = 50$ precomputed end-to-end paths over a horizon of $T = 5000$ time slots. At each slot, the network selects $L = 5$ paths from the available set $\mathcal{A}_t$ ($|\mathcal{A}_t| > L$) to serve concurrent connection requests. Results are averaged over 40 independent runs.

For each path i, the priority weight w_i is independently drawn from a uniform distribution over $(0, 1]$. Instantaneous rate, cost, and fidelity are sampled from truncated normal distributions with standard deviation $\sigma = 0.08$, truncated to $(0, 1)$. The fidelity threshold is set to $F_{\min}$, and the cost budget is $B = B_{\text{scale}} \cdot \frac{L}{K}\sum_{i=1}^{K}\mu_i$, where B_{scale} is a budget-scaling factor that controls the tightness of the cost constraint. Blocking delays follow a uniform distribution, with mean $d_i \sim \text{Unif}(0, 5)$ and $D_{\max} = \max_i 2d_i$; geometric and exponential distributions are also examined for robustness.

2) Baseline Algorithms: We implement the (α, β)-approximation oracle used in OPS-UCB via an FPTAS with approximation parameter $\epsilon = 0.01$, i.e., $\alpha = 0.99$ and $\beta = 1$. We compare OPS-UCB against the following baselines:

- **G-OPT**: An ideal full-information benchmark that uses the same approximation oracle with identical parameters as OQMPS. It has full knowledge of all paths parameters and select the feasible L paths that approximately maximize total weighted rates;
- **URM-UCB** (Unconstrained OQMPS): A variant of OPS-UCB which optimizes rate alone while ignoring cost and fidelity constraints;
- **EGD** (Heuristic Rate Delay Ratio Maximization): A heuristic that selects paths maximizing $\sum w_i\bar{r}_i/\bar{d}_i$ based on empirical averages. We consider this heuristic because of the blocking effect: a very high-rate path with an excessively long blocking delay may contribute less to the long-term average reward than a lower-rate path that is available more frequently. EGD therefore tries to balance immediate reward with frequent availability.

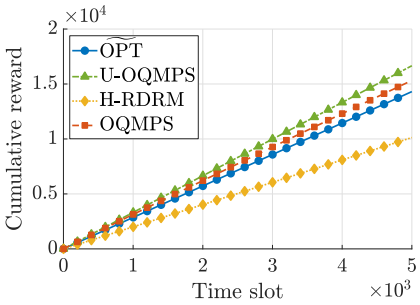
Performance is evaluated using three metrics: cumulative reward, regret against G-OPT, and total long-term violation.

B. Cumulative Reward and Regret

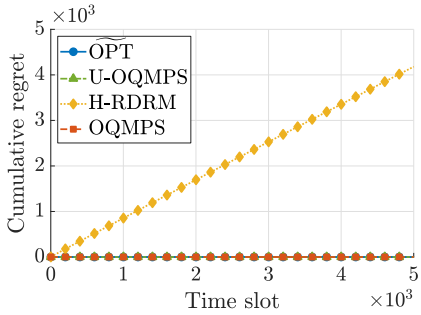
We first evaluate the four algorithms under the default configuration, where blocking delays follow a uniform distribution, the cost budget scaling factor is set to $B_{\text{scale}} = 1.0$, and the fidelity threshold is $F_{\min} = 0.7$.

Fig. 2a presents the cumulative reward over time. URM-UCB achieves the highest cumulative reward, as it greedily maximizes rate using UCB estimates while completely disregarding all constraints. Our proposed OPS-UCB attains the second-highest reward, significantly outperforming EGD. Notably, OPS-UCB surpasses the ideal benchmark G-OPT. This is because G-OPT operates under stricter constraint satisfaction in expectation, whereas OPS-UCB, being an online learning algorithm, adopts slightly more aggressive exploration that may incur marginal short-term violations. This tradeoff allows OPS-UCB to achieve higher realized reward than the ideal but conservative G-OPT benchmark without compromising long-term constraint safety.

Fig. 2b shows the cumulative regret relative to G-OPT. OPS-UCB achieves near-zero regret, indicating effective learning and convergence to near-optimal decisions under the given constraints. URM-UCB also exhibits a near-zero regret curve, but this does not imply convergence to a truly optimal feasible solution; rather, it reflects convergence to the unconstrained optimum that violates both cost and fidelity constraints, as evidenced by its severe violation in Fig. 3. In contrast,



(a) Cumulative Reward



(b) Cumulative Regret

Fig. 2: Comparison of cumulative reward and cumulative regret.

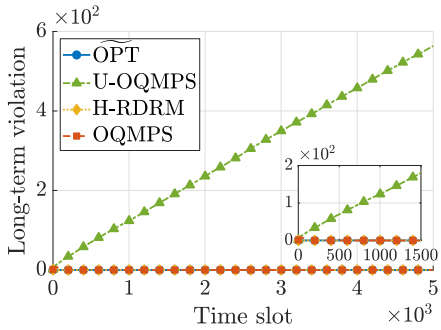


Fig. 3: Comparison of long-term violation.

EGD exhibits approximately linear regret growth. The inferior performance of EGD can be attributed to its objective of maximizing the weighted rate-to-delay ratio $w_i r_i/d_i$ rather than directly maximizing the rate itself.

1) Constraint Violation: Fig. 3 reports the total long-term violation, which accumulates the positive parts of both cost overruns and fidelity shortfalls. URM-UCB shows a continuously and nearly linear increase in violation over the entire horizon. This indicates that unconstrained rate maximization comes at the expense of resource budget exhaustion and frequent transmissions below the fidelity threshold.

In contrast, the violation of OPS-UCB remains near zero throughout the entire horizon, exhibiting only a negligible and extremely short-lived fluctuation during the very early exploration phase, which is an inherent consequence of the limited observations at the beginning. As the confidence bounds tighten with more observations, the algorithm successfully respects both constraints in the long run. EGD also exhibits near-zero violations in both metrics, despite its poor reward performance. This is because its empirical-mean-based decisions, while suboptimal for reward maximization, tend to be inherently conservative, thereby avoiding high-cost or low-fidelity paths.

These results collectively demonstrate that OPS-UCB achieves the highest reward among all constraint-aware algorithms while maintaining near-zero long-term violation, as guaranteed by our theoretical analysis in Section IV. The UCB-based optimistic estimation for rates enables effective exploration, while the LCB-based pessimistic estimation for costs provides robust budget protection, and the UCB-based filtering for fidelity ensures that only high-fidelity paths enter the candidate set.

2) Impact of Path Parameters: We then evaluate the robustness of OPS-UCB under varying configurations, focusing on three key parameters: the cost budget scaling factor B_{scale}, the fidelity threshold $F_{\min}$ and the delay distribution.

Impact of cost budget: We first vary the cost budget scaling factor $B_{\text{scale}} \in \{0.8, 0.9, 1.0, 1.1\}$. Fig. 4a and Fig. 4b report the cumulative reward and the total long-term violation, respectively.

As shown in Fig. 4a, the cumulative reward increases monotonically with B_{scale}. Notably, the performance gap across configurations remain moderate: even under the tightest budget, OPS-UCB attains substantial reward, indicating effective resource allocation. The inset figure confirms that the relative ordering stabilizes over time.

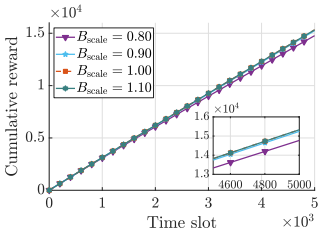
Fig. 4b reveals that the total long-term violation is highly sensitive to B_{scale}. For the tightest budget $B_{\text{scale}} = 0.8$, the violation rises persistently and significantly over the entire horizon, accumulating to the largest magnitude among all settings. When the budget is relaxed to $B_{\text{scale}} = 0.9$, the violation drops substantially compared with the 0.8 case, yet a non-negligible amount of cumulative violation still remains. In contrast, for $B_{\text{scale}} \geq 1.0$, the violation is effectively suppressed to near zero, with virtually no long-term accumulation. Critically, even under the tightest budget, the violation growth remains sublinear, consistent with our theoretical guarantee in Sec. IV.

It is worth noting that the total violation observed under tight budgets is primarily driven by cost overruns, as the fidelity constraint is largely decoupled from the cost budget: the UCB-based high-fidelity filtering mechanism operates independently, ensuring transmission quality even when the algorithm struggles to meet the cost limit. This is why the total violation for $B_{\text{scale}} = 0.8$ and 0.9 exhibits persistent accumulation—cost violations dominate—while for $B_{\text{scale}} \geq 1.0$, both cost and fidelity violations are simultaneously controlled.

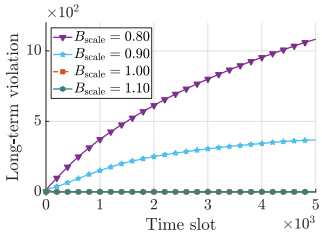
In summary, OPS-UCB maintains robust performance across varying budget settings: reward increases predictably with relaxed budgets, total constraint violation remains sublinear and vanish for $B_{\text{scale}} \geq 1.0$, and fidelity guarantees are preserved regardless of budget scaling.

Impact of fidelity threshold: We then vary the fidelity threshold $F_{\min} \in \{0.70, 0.75, 0.80\}$. Fig. 5a and Fig. 5b report the cumulative reward and the total long-term violation.

As shown in Fig. 5a, the cumulative reward decreases monotonically as $F_{\min}$ increases. This is because a stricter fidelity threshold shrinks the feasible set of paths that satisfy the constraint. Consequently, the algorithm has fewer options to select high-rate paths, leading to a lower overall reward. The strictest threshold yields the lowest reward, while the most

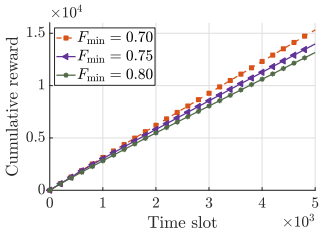


(a) Cumulative reward under different cost budget B

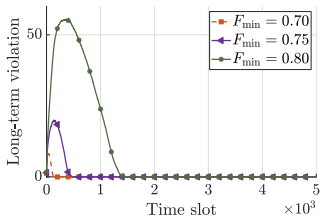


(b) Long-term violation under different cost budget B

Fig. 4: Performance under different cost budget.

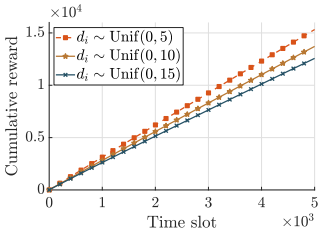


(a) Cumulative reward under different fidelity threshold F_{min}.

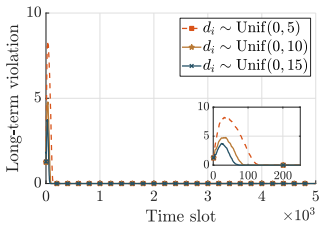


(b) Long-term violation under different fidelity threshold F_{min}.

Fig. 5: Performance under different fidelity threshold.

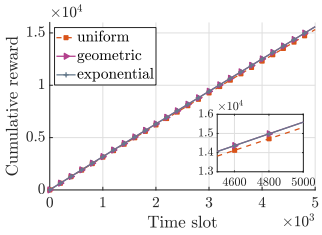


(a) Cumulative reward under different d_i.

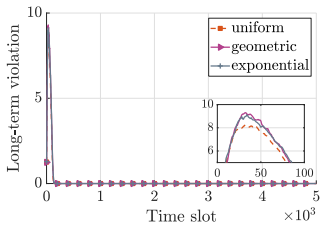


(b) Long-term violation under different d_i.

Fig. 6: Performance under different delay means.



(a) Cumulative reward under different delay distributions.



(b) Long-term violation under different delay distributions.

Fig. 7: Performance under different delay distributions.

lenient achieves the highest.

Fig. 5b presents the total long-term violation under different fidelity thresholds. The algorithm's behavior varies with the strictness of the threshold. For the most lenient threshold $F_{\min} = 0.70$, the violation remains virtually zero throughout the horizon, with no noticeable spike. As the threshold increases to $F_{\min} = 0.75$, a slight transient fluctuation appears during the early exploration phase but decays rapidly to zero. The strictest threshold $F_{\min} = 0.80$ exhibits a more visible initial spike and the slowest convergence. This inverse relationship stems from a simple fact: a higher threshold reduces the number of paths that naturally satisfy the fidelity requirement; consequently, regardless of the selection policy, any chosen set is more likely to include paths whose true fidelity falls below the threshold during the early learning phase, leading to larger and more persistent spikes. Crucially, for all three settings, the total long-term violation converges to zero, demonstrating the algorithm's robustness regardless of the fidelity requirement.

Impact of d_i: We investigate how the mean blocking delay d_i affects the performance of OPS-UCB by varying the upper bound of the uniform delay distribution, i.e., $d_i \sim \text{Unif}(0, 5)$, $\text{Unif}U(0, 10)$ and $\text{Unif}(0, 15)$.

As shown in Fig. 6a, the cumulative reward increases approximately linearly over time for all three settings, with a clear order: $d_i \sim \text{Unif}(0, 5)$ achieves the highest reward, followed by $d_i \sim \text{Unif}(0, 10)$, while $d_i \sim \text{Unif}(0, 15)$ yields the lowest. This is because shorter blocking delays allow paths with high optimistic rate estimates to be selected more frequently, thereby contributing more to the long-term average rate. The gap between the three curves widens gradually over time, and no convergence or crossover is observed.

Fig. 6b presents the total long-term violation under the same three delay settings. All violation curves remain extremely close to zero throughout the entire horizon, with no pronounced spikes or persistent accumulation. Moreover, the differences in violation across different d_i values are almost negligible. This indicates that the delay magnitude does not substantially affect the algorithm's ability to respect the constraints. The constraint enforcement mechanism of OPS-UCB is highly robust to the delay statistics, successfully maintaining a near-zero long-term violation regardless of the mean blocking delay.

3) Impact of blocking delay distributions: : We further test the robustness of OPS-UCB under different delay distributions, including uniform, geometric, and exponential distributions.

As shown in Fig. 7a, the cumulative reward is slightly higher under geometric and exponential delay distributions compared with the uniform case.

Fig. 7b presents the total long-term violation under the three delay distributions. Consistent with the results in Fig. 6b, the violation curves for all three distributions remain near zero throughout the entire horizon. There are no observable pronounced spikes or persistent accumulations, and the differences among the three curves are almost negligible. This demonstrates that the algorithm's constraint-satisfaction performance is highly robust to the shape of the delay distribution. Regardless of whether the delay is light-tailed or heavy-tailed, OPS-UCB successfully learns to respect both the cost and fidelity constraints in the long run.

VI. Conclusion & Future Work

This paper investigates the online multi-path selection problem in quantum networks, where an algorithm must learn unknown distribution of rate, cost, and fidelity while satisfying both a per-slot cost budget and a fidelity threshold. We propose an online learning algorithm that maintains confidence bounds for all path statistics, constructs optimistic estimates for rate and fidelity and a pessimistic estimate for cost, and invokes an (α, β)-approximation oracle to select feasible subset of paths. The theoretical analysis establishes sublinear ρ-approximate regret and sublinear total long-term violation. Simulations on a 50-path quantum network demonstrate that OPS-UCB achieves competitive reward compared to the ideal G-OPT benchmark while maintaining near-zero total long-term violation, validating the practical effectiveness of the approach.

Future work includes eliminating the reliance on precomputed end-to-end paths, extending the framework to accommodate multiple resource types, incorporating quantum memory coherence times explicitly into the blocking model, and exploring adaptive oracle precision tuning to further improve regret bounds in practical deployments.

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Appendix

A. Preliminaries for Regret Analysis

We first establish the structural properties of the optimal dynamic policy under stochastic blocking, and introduce the decomposition framework that connects the regret to the per-slot instantaneous loss.

Lemma 1. For any path i with expected blocking delay $d_i = \mathbb{E}[D_i(t)]$ and maximum delay $D_{\max}$, we have:

$$z_i \leq \frac{1}{d_i + 1} + \mathcal{O}\left(\frac{D_{\max}}{T}\right).$$

Proof. We denote by $\mathcal{P}_t^*$ and $\mathcal{B}_t^*$ the set of selected and blocked paths of optimal online policy π^*, respectively, at any time t. The event that path i is selected at time slot t and the event that i is blocked at t are mutually exclusive, namely, $\mathbb{I}(i \in \mathcal{P}_t^*) + \mathbb{I}(i \in \mathcal{B}_t^*) \leq 1$. Further, an arm i is blocked at t if and only if the arm has been played in some previous time slot and is still unavailable at t. Therefore, at any time slot t, we get:

$$\mathbb{I}(i \in \mathcal{P}_t^*) + \sum_{t' < t} \mathbb{I}(i \in \mathcal{P}_{t'}^*)\mathbb{I}(D_i(t') \geq t - t') \leq 1.$$

By taking expectation in the above expression, we have that:

$$\mathbb{P}(i \in \mathcal{P}_t^*) + \sum_{t' < t} \mathbb{E}[\mathbb{I}(i \in \mathcal{P}_{t'}^*)\mathbb{I}(D_i(t') \geq t - t')] \leq 1, \ \forall t \tag{10}$$

Since the events $\{i \in \mathcal{P}_t^*\}$ and $\{D_i(t') \geq t - t'\}$ are conditionally independent given the history of paths selected up to (and including) $t' - 1$. We denote the history up to t as H_t^*, then:

$$\begin{aligned}
&\mathbb{E}[\mathbb{I}(i \in \mathcal{P}_{t'}^*)\mathbb{I}(D_i(t') \geq t - t')] \\
&= \mathbb{E}\big[\mathbb{E}[\mathbb{I}(i \in \mathcal{P}_{t'}^*)\mathbb{I}(D_i(t') \geq t - t') \mid H_{t'-1}^*]\big] \\
&= \mathbb{E}\big[\mathbb{E}[\mathbb{I}(i \in \mathcal{P}_{t'}^*) \mid H_{t'-1}^*]\mathbb{E}[\mathbb{I}(D_i(t') \geq t - t') \mid H_{t'-1}^*]\big] \\
&= \mathbb{P}(i \in \mathcal{P}_{t'}^*)\mathbb{P}(D_i(t') \geq t - t')
\end{aligned}$$

By summing (10) over all time slots t and since $\mathbb{P}(D_i(t) \geq 0) = 1$, we get:

$$\sum_{t=1}^{T} \sum_{t' \leq t} \mathbb{P}(i \in \mathcal{P}_{t'}^*)\mathbb{P}(D_i(t') \geq t - t') \leq T. \tag{11}$$

Since the variables $D_i(t)$ are i.i.d., we can consider any sample D_i from the delay distribution of path i and have $\mathbb{P}(D_i \geq t - t') = \mathbb{P}(D_i(t) \geq t - t')$. Moreover, since D_i has support in $\{0, 1, ..., D_{\max}\}$, we can write:

$$\begin{aligned}
&\sum_{t=1}^{T} \sum_{t' \leq t} \mathbb{P}(i \in \mathcal{P}_{t'}^*)\mathbb{P}(D_i(t') \geq t - t') \\
&= \sum_{t=1}^{T} \sum_{t'=\max\{1, t - D_{max}\}}^{t} \mathbb{P}(i \in \mathcal{P}_{t'}^*)\mathbb{P}(D_i \geq t - t') \\
&= \sum_{t'=1}^{T} \sum_{t=t'}^{\min\{T, t'+D_{\max}\}} \mathbb{P}(i \in \mathcal{P}_{t'}^*)\mathbb{P}(D_i \geq t - t') \\
&= \sum_{t'=1}^{T} \mathbb{P}(i \in \mathcal{P}_{t'}^*) \sum_{t=t'}^{\min\{T, t'+D_{\max}\}} \mathbb{P}(D_i \geq t - t') \\
&\geq \sum_{t'=1}^{T} \mathbb{P}(i \in \mathcal{P}_{t'}^*) \sum_{t=t'}^{t'+D_{\max}} \mathbb{P}(D_i \geq t - t') \\
&\quad - \sum_{t'=T-D_{\max}+1}^{T} \sum_{t=t'}^{T} \mathbb{P}(D_i \geq t - t') \\
&= \sum_{t'=1}^{T} \mathbb{P}(i \in \mathcal{P}_{t'}^*)(1 + \mathbb{E}[D_i]) \\
&\quad - \sum_{t'=T-D_{\max}+1}^{T} \sum_{t=t'}^{T} \mathbb{P}(D_i \geq t - t'),
\end{aligned}$$

Therefore, inequality (11) becomes:

$$\begin{aligned}
&\sum_{t'=1}^{T} \mathbb{P}(i \in \mathcal{P}_{t'}^*)(1 + \mathbb{E}[D_i]) \\
&\quad - \sum_{t'=T-D_{\max}+1}^{T} \sum_{t=t'}^{T} \mathbb{P}(D_i \geq t - t') \leq T.
\end{aligned}$$

Finally, by rearranging terms in the above expression and dividing by $(\mathbb{E}[D_i] + 1) \cdot T$, we conclude that:

$$\begin{aligned}
\frac{1}{T} \sum_{t'=1}^{T} \mathbb{P}(i \in \mathcal{P}_{t'}^*) &\leq \frac{1}{(\mathbb{E}[D_i] + 1)} \\
&\quad + \frac{1}{T(\mathbb{E}[D_i] + 1)} \sum_{t'=T-D_{\max}+1}^{T} \sum_{t=t'}^{T} \mathbb{P}(D_i \geq t - t') \\
&\leq \frac{1}{(\mathbb{E}[D_i] + 1)} + \frac{D_{\max}(\mathbb{E}[D_i] + 1)}{T(\mathbb{E}[D_i] + 1)} \\
&= \frac{1}{(\mathbb{E}[D_i] + 1)} + \frac{D_{\max}}{T}.
\end{aligned}$$

□

Lemma 2. For any subset of paths $\mathcal{S} \subseteq [K]$, let $\mathfrak{P}(\mathcal{S})$ be the feasible family defined in Definition 1, then we have:

$$\max_{\mathcal{Q} \in \mathfrak{P}(\mathcal{S})} \sum_{i \in \mathcal{Q}} w_i r_i \geq \sum_{i \in \mathcal{S}} w_i r_i z_i$$

Proof. At any time t, we have:

$$\max_{\mathcal{Q} \subseteq \mathcal{S}} \sum_{i \in \mathcal{Q}} w_i r_i \geq \sum_{i \in \mathcal{S} \cap \mathcal{P}_{t'}^*} w_i r_i = \sum_{i \in \mathcal{S}} w_i r_i \mathbb{I}(i \in \mathcal{P}_{t'}^*).$$

By averaging over time and taking expectation over t, we can conclude that:

$$\max_{\mathcal{Q}\subseteq\mathcal{S}} \sum_{i\in\mathcal{Q}} w_i r_i \geq \frac{1}{T}\sum_{t=1}^{T}\sum_{i\in\mathcal{S}} w_i r_i \mathbb{P}(i\in\mathcal{P}_{t'}^*) = \sum_{i\in\mathcal{S}} w_i r_i z_i.$$

□

Definition 2 (Instantaneous regret)**.** At time t, the instantaneous regret of the algorithm is defined as:

$$\Gamma_t = \alpha\beta \cdot \max_{\mathcal{Q}\in\mathfrak{P}(\mathcal{A}_t)} \sum_{i\in\mathcal{Q}} w_i r_i - \sum_{i\in\mathcal{P}_t} w_i r_i,$$

where $\mathcal{A}_t$ is the available paths at the beginning of time slot t, $\mathfrak{P}(\mathcal{A}_t)$ is the family of feasible subsets of $\mathcal{A}_t$, and $\mathcal{P}_t$ is the set selected by the algorithm. Intuitively, Γ_t measures the gap between an $\alpha\beta$-fraction of the optimal value over the available paths and the actual reward obtained.

Lemma 3. The ρ-approximate regret of the algorithm, with $\rho = \frac{\alpha\beta}{1+\alpha\beta}$, satisfies:

$$\mathrm{Reg}_\rho(T) \leq \frac{1}{1+\alpha\beta}\mathbb{E}\left[\sum_{t=1}^{T}\Gamma_t\right] + \mathcal{O}(D_{\max}\cdot K).$$

Proof. Let H_t be the history of path selecting of our bandit algorithm up to (including) time t. We define the total expected reward in T time slots by our algorithm as:

$$\begin{aligned}
\left[\sum_{t=1}^{T} r(t)\right] &= \mathbb{E}\left[\sum_{t=1}^{T}\sum_{i=1}^{K} w_i r_i(t)\mathbb{I}(i\in\mathcal{P}_t)\right] \\
&= \sum_{t=1}^{T}\sum_{i=1}^{K}\mathbb{E}[w_i r_i(t)\mathbb{I}(i\in\mathcal{P}_t] \\
&= \sum_{t=1}^{T}\sum_{i=1}^{K}\mathbb{E}\big[\mathbb{E}[w_i r_i(t)\mathbb{I}(i\in\mathcal{P}_t)|H_{t-1}]\big] \\
&= \sum_{t=1}^{T}\sum_{i=1}^{K}\mathbb{E}\big[\mathbb{E}[w_i r_i(t)|H_{t-1}]\mathbb{E}[\mathbb{I}(i\in\mathcal{P}_t)|H_{t-1}]\big] \\
&= \sum_{t=1}^{T}\sum_{i=1}^{K}\mathbb{E}[w_i r_i\mathbb{I}(i\in\mathcal{P}_t)] \\
&= \sum_{t=1}^{T}\mathbb{E}\left[\sum_{i\in\mathcal{P}_t} w_i r_i\right],
\end{aligned}$$

where we use the fact that, at each time slot t, the reward $w_i r_i$ and the choice of the set $\mathcal{P}_t$ are independent conditioned on the history H_{t-1}.

We recall the expected pulling rate of each arm:

$$z_i = \mathbb{E}\left[\frac{1}{T}\sum_{t=1}^{T}\mathbb{I}(i\in\mathcal{P}_t^*)\right]$$

where $\mathcal{P}_t^*$ is the set of paths selected by the optimal online policy π^* at t.

A direct application of the definition of Γ_t, for every t, gives:

$$\sum_{i\in\mathcal{P}_t} w_i r_i = \alpha\beta\cdot\max_{\mathcal{Q}\subseteq\mathcal{A}_t}\sum_{i\in\mathcal{Q}} w_i r_i - \Gamma_t.$$

where $\mathcal{A}_t$ is the available paths of our algorithm at the beginning of time slot t, $\mathcal{Q}$ is the feasible subset satisfying the cardinality, cost, and fidelity constraint.

We can see that for any t:

$$\begin{aligned}
\mathbb{E}\left[\sum_{i\in\mathcal{P}_t} w_i r_i\right] &= \alpha\beta\cdot\mathbb{E}\left[\max_{\mathcal{Q}\subseteq\mathcal{A}_t}\sum_{i\in\mathcal{Q}} w_i r_i\right] - \mathbb{E}[\Gamma_t] \\
&\geq \alpha\beta\cdot\mathbb{E}\left[\sum_{i\in\mathcal{A}_t} w_i r_i z_i\right] - \mathbb{E}[\Gamma_t] \\
&= \alpha\beta\cdot\sum_{i=1}^{K} w_i r_i z_i - \alpha\beta\cdot\sum_{i\in\mathcal{B}_t} w_i r_i z_i - \mathbb{E}[\Gamma_t]
\end{aligned}$$

where $\mathcal{B}_t$ is the blocked (unavailable) paths of our algorithm. The inequality above is due to Lemma 2.

Note that for any t and arm $i\in\mathcal{B}_t$, we have:

$$\mathbb{I}(i\in\mathcal{B}_t) = \sum_{t'<t}\mathbb{I}(i\in\mathcal{P}_{t'})\mathbb{I}(D_i(t')\geq t-t').$$

Using this fact, we have:

$$\begin{aligned}
&\mathbb{E}\left[\sum_{t=1}^{T}\sum_{i\in\mathcal{B}_t} w_i r_i z_i\right] \\
&= \mathbb{E}\left[\sum_{t=1}^{T}\sum_{i=1}^{K} w_i r_i z_i \sum_{t'<t}\mathbb{I}(i\in\mathcal{P}_{t'})\mathbb{I}(D_i(t')\geq t-t')\right] \\
&= \sum_{t=1}^{T}\sum_{i=1}^{K} w_i r_i \sum_{t'<t}\mathbb{P}(i\in\mathcal{P}_{t'})\mathbb{P}(D_i(t')\geq t-t') \\
&= \sum_{t'=1}^{T}\sum_{i=1}^{K}\mathbb{P}(i\in\mathcal{P}_{t'}) w_i r_i z_i \sum_{t>t'}\mathbb{P}(D_i(t')\geq t-t') \\
&\leq \sum_{t'=1}^{T}\sum_{i=1}^{K}\mathbb{P}(i\in\mathcal{P}_{t'}) w_i r_i z_i d_i \\
&\leq \left[\sum_{t=1}^{T} r(t)\right] + \mathcal{O}(D_{\max}\cdot K)
\end{aligned} \tag{12}$$

By summing over t, we have:

$$\begin{aligned}
&\left[\sum_{t=1}^{T} r(t)\right] = \mathbb{E}\left[\sum_{t=1}^{T}\sum_{i=1}^{K} w_i r_i(t)\mathbb{I}(i\in\mathcal{P}_t)\right] \\
&= \sum_{t=1}^{T}\mathbb{E}\left[\sum_{i\in\mathcal{P}_t} w_i r_i\right] \\
&\geq \alpha\beta\mathrm{Rew}^*(T) - \alpha\beta\left[\sum_{t=1}^{T} r(t)\right] - \mathcal{O}(D_{\max}\cdot K) - \mathbb{E}\left[\sum_{t=1}^{T}\Gamma_t\right].
\end{aligned}$$

By rearranging terms in the expression above, we get that:

$$\left[\sum_{t=1}^{T} r(t)\right] \geq \frac{\alpha\beta}{1+\alpha\beta}\mathrm{Rew}^*(T) - \mathcal{O}(D_{\max}\cdot K) - \frac{1}{1+\alpha\beta}\mathbb{E}\left[\sum_{t=1}^{T}\Gamma_t\right]$$

□

Lemma 4. Let $\mathcal{N}_t$ denote the event that for all paths $i \in [K]$, the following inequalities hold simultaneously: $|\bar{r}_i(t-1) - r_i| \leq \mathrm{rad}_i(t)$, $|\bar{\mu}_i(t-1) - \mu_i| \leq \mathrm{rad}_i(t)$, $|\bar{F}_i(t-1) - F_i| \leq \mathrm{rad}_i(t)$, where $\mathrm{rad}_i(t) = \sqrt{\frac{3\ln t}{2n_i(t-1)}}$. Then we have $\mathbb{P}(\mathcal{N}_t) \geq 1 - \frac{6K}{t^3}$.

Proof. For any $n_i(t-1) > 0$ and $i \in [K]$, using Hoeffding's inequality and the fact that $n_i(t-1) \leq t-1$, we have that:

$$\begin{aligned}
&\mathbb{P}\left(|\bar{r}_i(t-1) - r_i| \geq \sqrt{\frac{3\ln t}{2n_i(t-1)}}\right) \\
&= \sum_{s=1}^{t-1}\mathbb{P}(n_i(t-1)=s)\cdot\mathbb{P}\left(|\bar{r}_i(t-1)-r_i| \geq \sqrt{\frac{3\ln t}{2s}}\,\middle|\, n_i(t-1)=s\right) \\
&\leq \sum_{s=1}^{t-1}\mathbb{P}(n_i(t-1)=s)\cdot 2e^{-3\ln t} \\
&= 2t^{-3}\sum_{s=1}^{t-1}\mathbb{P}(n_i(t-1)=s) \\
&\leq \frac{2}{t^3}
\end{aligned}$$

We also have that:

$$\mathbb{P}\left(|\bar{\mu}_i(t-1) - \mu_i| \geq \sqrt{\frac{3\ln t}{2n_i(t-1)}}\right) \leq \frac{2}{t^3}$$
$$\mathbb{P}\left(|\bar{F}_i(t-1) - F_i| \geq \sqrt{\frac{3\ln t}{2n_i(t-1)}}\right) \leq \frac{2}{t^3}$$

The lemma follows by taking union bound over the three variables and over all K paths:

$$\begin{aligned}
&\mathbb{P}(\neg\mathcal{N}_t) = \mathbb{P}\Big(\exists i \in [K] : |\bar{r}_i(t-1) - r_i| > \mathrm{rad}_i(t) \text{ or } \\
&|\bar{\mu}_i(t-1) - \mu_i| > \mathrm{rad}_i(t) \text{ or } |\bar{F}_i(t-1) - F_i| > \mathrm{rad}_i(t)\Big) \\
&\leq 3\cdot\sum_{i=1}^{K}\frac{2}{t^3} \leq \frac{6K}{t^3}.
\end{aligned}$$

□

B. Proof of Theorem 1

With the above definitions and lemmas in place, we now present the complete proof of the regret bound.

Theorem 1. For the online algorithm with an (α, β)-approximation oracle, the ρ-approximate regret, with $\rho = \frac{\alpha\beta}{1+\alpha\beta}$, satisfies the following distribution-independent bound:

$$\mathrm{Reg}_\rho(T) \leq \frac{14\sqrt{KLT\ln T}}{1+\alpha\beta} + \frac{K}{1+\alpha\beta}(2+\pi^2 L) + \mathcal{O}(D_{\max}\cdot K).$$

where $\Delta_{\max}$ denotes the maximum possible suboptimality gap described in Appendix.

Proof. According to Lemma 3, to bound the ρ-approximate regret of our bandit algorithm, it suffices to bound the loss accumulated due to the instantaneous regret over time. This loss can be written as:

$$\begin{aligned}
\mathbb{E}\left[\sum_{t=1}^{T}\Gamma_t\right] &= \mathbb{E}\left[\sum_{t=1}^{T}\left(\alpha\beta\cdot\max_{\mathcal{Q}\subseteq\mathcal{A}_t}\sum_{i\in\mathcal{Q}} w_i r_i - \sum_{i\in\mathcal{P}_t} w_i r_i\right)\right] \\
&= \mathbb{E}\Bigg[\sum_{t=1}^{T}\Bigg(\alpha\cdot\max_{\mathcal{Q}\subseteq\mathcal{A}_t}\sum_{i\in\mathcal{Q}} w_i r_i - \sum_{i\in\mathcal{P}_t} w_i r_i \\
&\qquad + \alpha(\beta-1)\cdot\max_{\mathcal{Q}\subseteq\mathcal{A}_t}\sum_{i\in\mathcal{Q}} w_i r_i\Bigg)\Bigg].
\end{aligned}$$

Focusing on the term $\alpha\cdot\max_{\mathcal{Q}\subseteq\mathcal{A}_t}\sum_{i\in\mathcal{Q}} w_i r_i - \sum_{i\in\mathcal{P}_t} w_i r_i$ of the loss, we define bad set and suboptimality gap to characterize the decision that contributes to this quantity:

Definition 3 (Bad set)**.** A subset $\mathcal{S} \subseteq \mathcal{A}$ is called a bad set with respect to the available set $\mathcal{A}$ if: $\sum_{i\in\mathcal{S}} w_i r_i \leq \alpha\cdot\max_{\mathcal{Q}\subseteq\mathcal{A}}\sum_{i\in\mathcal{Q}} w_i r_i$. Moreover, we define the bad sets of available set $\mathcal{A}$ as:

$$\mathcal{S}_B(\mathcal{A}) = \left\{\mathcal{S}\;\middle|\;\sum_{i\in\mathcal{S}} w_i r_i \leq \alpha\cdot\max_{\mathcal{Q}\subseteq\mathcal{A}}\sum_{i\in\mathcal{Q}} w_i r_i\right\}$$

where $|\mathcal{S}| = L$, and $\mathcal{Q}$ is the feasible subset satisfying the cardinality, cost, and fidelity constraints.

Definition 4 (Suboptimality gap)**.** The suboptimality gap of a bad set $\mathcal{S}$ with respect to $\mathcal{A}$ is defined as:

$$\Delta(\mathcal{S},\mathcal{A}) = \alpha\cdot\max_{\mathcal{Q}\subseteq\mathcal{A}}\sum_{i\in\mathcal{Q}} w_i r_i - \sum_{i\in\mathcal{S}} w_i r_i.$$

where $|\mathcal{S}| = L$, and $\mathcal{Q}$ is the feasible subset satisfying the cardinality, cost, and fidelity constraints.

Let $\Delta_{\max} = \max_{\mathcal{S}\in\mathcal{S}_B(\mathcal{A}),\mathcal{A}\subseteq\{1,\ldots,K\}}\Delta(\mathcal{S},\mathcal{A})$ denote the maximum possible gap.

Thus, we can write:

$$\begin{aligned}\mathbb{E}\left[\sum_{t=1}^{T}\Gamma_t\right] &= \mathbb{E}\left[\sum_{t=1}^{T}\left(\alpha\cdot\max_{\mathcal{Q}\subseteq\mathcal{A}_t}\sum_{i\in\mathcal{Q}}w_i r_i - \sum_{i\in\mathcal{P}_t}w_i r_i \right.\right.\\ &\quad\left.\left.+\alpha(\beta-1)\cdot\max_{\mathcal{Q}\subseteq\mathcal{A}_t}\sum_{i\in\mathcal{Q}}w_i r_i\right)\right]\\ &\leq \mathbb{E}\left[\sum_{t=1}^{T}\left(\mathbb{I}(\mathcal{P}_t\in\mathcal{S}_B(\mathcal{A}_t))(\alpha\cdot\max_{\mathcal{Q}\subseteq\mathcal{A}_t}\sum_{i\in\mathcal{Q}}w_i r_i\right.\right.\\ &\quad\left.\left.-\sum_{i\in\mathcal{P}_t}w_i r_i)+\alpha(\beta-1)\cdot\max_{\mathcal{Q}\subseteq\mathcal{A}_t}\sum_{i\in\mathcal{Q}}w_i r_i\right)\right]\\ &= \mathbb{E}\left[\sum_{t=1}^{T}\left(\mathbb{I}(\mathcal{P}_t\in\mathcal{S}_B(\mathcal{A}_t))\Delta(\mathcal{P}_t,\mathcal{A}_t)\right.\right.\\ &\quad\left.\left.+\alpha(\beta-1)\cdot\max_{\mathcal{Q}\subseteq\mathcal{A}_t}\sum_{i\in\mathcal{Q}}w_i r_i\right)\right]\end{aligned} \tag{13}$$

Now we can treat the parts of the regret that result from oracle fails or non-representative sampling separately. Let $\mathcal{O}_t$ denote the event such that oracle successes at time slot t. We decompose the indicator in (13) based on whether the events $\mathcal{O}_t$ and $\mathcal{N}_t$ hold, as follows:

$$(13) \leq \mathbb{E}\left[\sum_{t=1}^{T}\mathbb{I}(\mathcal{O}_t,\mathcal{N}_t,\mathcal{P}_t\in\mathcal{S}_B(\mathcal{A}_t))\cdot\Delta(\mathcal{P}_t,\mathcal{A}_t)\right] \tag{14}$$

$$+\mathbb{E}\left[\sum_{t=1}^{T}\mathbb{I}(\neg\mathcal{O}_t,\mathcal{P}_t\in\mathcal{S}_B(\mathcal{A}_t))\cdot\Delta(\mathcal{P}_t,\mathcal{A}_t)+\alpha(\beta-1)\cdot\max_{\mathcal{Q}\subseteq\mathcal{A}_t}\sum_{i\in\mathcal{Q}}w_i r_i\right] \tag{15}$$

$$+\mathbb{E}\left[\sum_{t=1}^{T}\mathbb{I}(\neg\mathcal{N}_t,\mathcal{P}_t\in\mathcal{S}_B(\mathcal{A}_t))\cdot\Delta(\mathcal{P}_t,\mathcal{A}_t)\right] \tag{16}$$

We bound each one of the above terms separately. For (15), to address the issue that the availability set changes at each round, we consider the suboptimality gap of the worst solution contained in the availability set $\mathcal{F}$, $\Delta_{\max}(\mathcal{F})$, namely:

$$\Delta_{\max}(\mathcal{F}) = \max_{\mathcal{S}\in\mathcal{S}_B}\Delta(\mathcal{S},\mathcal{F}).$$

We use the above definition to upper bound the suboptimality gap appearing in (15). We have that:

$$\begin{aligned}(15) &\leq \mathbb{E}\left[\sum_{t=1}^{T}\mathbb{I}\Big((\neg\mathcal{O}_t)\cdot\Delta_{\max}(\mathcal{A}_t)+\alpha(\beta-1)\cdot\max_{\mathcal{Q}\subseteq\mathcal{A}_t}\sum_{i\in\mathcal{Q}}w_i r_i\Big)\right]\\ &= \mathbb{E}\left[\sum_{t=1}^{T}\sum_{\mathcal{F}\subseteq[K]}\mathbb{E}\Big[\mathbb{I}(\neg\mathcal{O}_t)\cdot\Delta_{\max}(\mathcal{F})+\alpha(\beta-1)\cdot\max_{\mathcal{Q}\subseteq\mathcal{A}_t}\sum_{i\in\mathcal{Q}}w_i r_i\Big|\mathcal{A}_t=\mathcal{F}\Big]\mathbb{I}(\mathcal{A}_t=\mathcal{F})\right]\\ &= \mathbb{E}\left[\sum_{t=1}^{T}\sum_{\mathcal{F}\subseteq[K]}\Big(\mathbb{P}(\neg\mathcal{O}_t|\mathcal{A}_t=\mathcal{F})\cdot\Delta_{\max}(\mathcal{F})+\alpha(\beta-1)\cdot\max_{\mathcal{Q}\subseteq\mathcal{A}_t}\sum_{i\in\mathcal{Q}}w_i r_i\Big)\mathbb{I}(\mathcal{A}_t=\mathcal{F})\right]\\ &\leq \mathbb{E}\left[\sum_{t=1}^{T}\sum_{\mathcal{F}\subseteq[K]}\Big((1-\beta)\cdot\Delta_{\max}(\mathcal{F})+\alpha(\beta-1)\cdot\max_{\mathcal{Q}\subseteq\mathcal{A}_t}\sum_{i\in\mathcal{Q}}w_i r_i\Big)\mathbb{I}(\mathcal{A}_t=\mathcal{F})\right]\leq 0\end{aligned} \tag{17}$$

For the term (16), using the definition of $\Delta_{\max}$ and Lemma 4, we have that:

$$\begin{aligned}(16) &= \mathbb{E}\left[\sum_{t=1}^{T}\mathbb{I}(\neg\mathcal{N}_t)\cdot\Delta_{\max}\right] = \sum_{t=1}^{T}\mathbb{P}(\neg\mathcal{N}_t)\cdot\Delta_{\max}\\ &\leq \sum_{t=1}^{T}\frac{6K}{t^3}\Delta_{\max}\leq\sum_{t=1}^{T}\frac{6K}{t^2}\Delta_{\max}\leq\pi^2K\Delta_{\max}\leq\pi^2KL.\end{aligned} \tag{18}$$

To bound (14), we first define the sampling threshold for a given gap level M:

Definition 5 (Sampling threshold)**.** We define the sampling threshold as:

$$l_T(M) = \frac{24L^2\ln T}{M^2},$$

We also define the following function:

$$\kappa_T(M,s) = \begin{cases}2, & s=0,\\ \sqrt{\frac{24\ln T}{s}}, & 1\leq s\leq l_T(M),\\ 0, & s>l_T(M).\end{cases}$$

We define $M=\sqrt{(48KL\ln T)/T}$. We use this definition

to rewrite the term (14) as:

$$(14) = \mathbb{E}\Bigg[\sum_{t=1}^{T}\mathbb{I}(\mathcal{O}_t,\mathcal{N}_t,\mathcal{P}_t\in\mathcal{S}_B(\mathcal{A}_t),\Delta(\mathcal{P}_t,\mathcal{A}_t)\geq M) \cdot \Delta(\mathcal{P}_t,\mathcal{A}_t)\Bigg] \tag{19}$$

$$+\mathbb{E}\Bigg[\sum_{t=1}^{T}\mathbb{I}(\mathcal{O}_t,\mathcal{N}_t,\mathcal{P}_t\in\mathcal{S}_B(\mathcal{A}_t),\Delta(\mathcal{P}_t,\mathcal{A}_t)< M) \cdot \Delta(\mathcal{P}_t,\mathcal{A}_t)\Bigg]. \tag{20}$$

The second term of the above equation can be bounded using the fact that $\Delta(\mathcal{P}_t,\mathcal{A}_t) < M$, as follows:

$$\begin{aligned}(20) &\leq \mathbb{E}\Bigg[\sum_{t=1}^{T}\mathbb{I}(\mathcal{P}_t\in\mathcal{S}_B(\mathcal{A}_t),\Delta(\mathcal{P}_t,\mathcal{A}_t)< M) \cdot \Delta(\mathcal{P}_t,\mathcal{A}_t)\Bigg] \leq T\cdot M = T\sqrt{(48KL\ln T)/T} \\ &= \sqrt{48LKT\ln T}.\end{aligned} \tag{21}$$

The first term can be bounded using Lemma 5:

$$\begin{aligned}(19) &= \mathbb{E}\Bigg[\sum_{t=1}^{T}\mathbb{I}(\mathcal{O}_t,\mathcal{N}_t,\mathcal{P}_t\in\mathcal{S}_B(\mathcal{A}_t),\Delta(\mathcal{P}_t,\mathcal{A}_t)\geq M) \cdot \Delta(\mathcal{P}_t,\mathcal{A}_t)\Bigg] \\ &\leq \mathbb{E}\Bigg[\sum_{t=1}^{T}\mathbb{I}(\mathcal{O}_t,\mathcal{N}_t,\mathcal{P}_t\in\mathcal{S}_B(\mathcal{A}_t),\Delta(\mathcal{P}_t,\mathcal{A}_t)\geq M) \cdot \kappa_T(M,n_i(t-1))\Bigg] \\ &\leq \mathbb{E}\Bigg[\sum_{i=1}^{K}\sum_{s=0}^{n_i(T)}\kappa_T(M,s)\Bigg] \\ &\leq 2K + \sum_{i=1}^{K} 48\frac{L}{M}\ln T \\ &= 2K + \sqrt{48KLT\ln T}\end{aligned} \tag{22}$$

We conclude our proof by combining the above (17), (18), (21) and (22) with Lemma 3:

$$\begin{aligned}\mathrm{Reg}_\rho(T) &\leq \frac{1}{1+\alpha\beta}\mathbb{E}\Bigg[\sum_{t=1}^{T}\Gamma_t\Bigg] + \mathcal{O}(D_{\max}\cdot K) \\ &\leq \frac{1}{1+\alpha\beta}\Bigg(\pi^2 KL + 2\sqrt{48KLT\ln T} + 2K\Bigg) + \mathcal{O}(D_{\max}\cdot K) \\ &\leq \frac{14\sqrt{KLT\ln T}}{1+\alpha\beta} + \frac{K}{1+\alpha\beta}(2+\pi^2 L) + \mathcal{O}(D_{\max}\cdot K).\end{aligned}$$

□

Lemma 5. For any time t, if both $(\mathcal{O}_t)$ and $(\mathcal{N}_t)$ hold, the algorithm selects a bad set $\mathcal{P}_t \in \mathcal{S}_B(\mathcal{A}_t)$, and $\Delta(\mathcal{P}_t,\mathcal{A}_t) \geq M$, then:

$$\Delta(\mathcal{P}_t,\mathcal{A}_t) \leq \sum_{i\in\mathcal{A}_t}\kappa_T(M,n_i(t-1)).$$

Proof. By definition of $\mathcal{N}_t$, at any time slot t, if $\mathcal{N}_t$ holds, then $\forall i$ the optimistic estimates and pessimistic estimate at t can be bounded as follows:

$$|\hat{r}_i(t) - r_i| \leq \min\left\{2\sqrt{\frac{3\ln t}{2n_i(t-1)}}, 1\right\}, \tag{23}$$

$$|\check{\mu}_i(t) - \mu_i| \leq \left\{2\sqrt{\frac{3\ln t}{2n_i(t-1)}}, 1\right\}, \tag{24}$$

$$|\hat{F}_i(t) - F_i| \leq \left\{2\sqrt{\frac{3\ln t}{2n_i(t-1)}}, 1\right\}, \tag{25}$$

and:

$$\hat{r}_i(t) \geq r_i. \tag{26}$$

$$\check{\mu}_i(t) \leq \mu_i. \tag{27}$$

$$\hat{F}_i(t) \geq F_i. \tag{28}$$

Then, if $\mathcal{O}_t$ holds, we have that:

$$\begin{aligned}\sum_{i\in\mathcal{P}_t} w_i\hat{r}_i(t) &\geq \alpha\cdot\max_{\mathcal{Q}'\subseteq\mathcal{A}_t}\sum_{i\in\mathcal{Q}'} w_i\hat{r}_i(t) \geq \alpha\cdot\max_{\mathcal{Q}\subseteq\mathcal{A}_t}\sum_{i\in\mathcal{Q}} w_i\hat{r}_i(t) \\ &\geq \alpha\cdot\max_{\mathcal{Q}\subseteq\mathcal{A}_t}\sum_{i\in\mathcal{Q}} w_i r_i = \sum_{i\in\mathcal{P}_t} w_i r_i + \Delta(\mathcal{P}_t,\mathcal{A}_t),\end{aligned}$$

where $\mathcal{Q}' \subseteq \mathcal{A}_t$ is a subset satisfying $|\mathcal{Q}'| = L$, $\sum_{i\in\mathcal{Q}'}\check{\mu}_i(t) \leq B$ and $\hat{F}_i(t) \geq F_{\min}, \forall i \in \mathcal{Q}'$, and $\mathcal{Q} \subseteq \mathcal{A}_t$ is a real feasible subset satisfying $|\mathcal{Q}| = L$, $\sum_{i\in\mathcal{Q}}\mu_i \leq B$ and $F_i \geq F_{\min}, \forall i \in \mathcal{Q}$. When $\mathcal{N}_t$ holds, using (27) and (28), we have $\mathcal{Q} \subseteq \mathcal{Q}'$ as:

$$\sum_{i\in\mathcal{Q}}\check{\mu}_i(t) \leq \sum_{i\in\mathcal{Q}}\mu_i \leq B,$$

$$\hat{F}_i(t) \geq F_i \geq F_{\min}, \forall i \in \mathcal{Q}.$$

By reordering terms, we get that:

$$\begin{aligned}\Delta(\mathcal{P}_t,\mathcal{A}_t) &\leq \sum_{i\in\mathcal{P}_t} w_i\hat{r}_i(t) - \sum_{i\in\mathcal{P}_t} w_i r_i \\ &\leq \sum_{i\in\mathcal{P}_t} w_i|\hat{r}_i(t) - r_i|.\end{aligned} \tag{29}$$

From (29) and the fact that $\Delta(\mathcal{P}_t,\mathcal{A}_t) \geq M$ we obtain: $\Delta(\mathcal{P}_t,\mathcal{A}_t) \leq 2\sum_{i\in\mathcal{P}_t} w_i|\hat{r}_i(t) - r_i| - M$. This can be rewritten as:

$$\begin{aligned}\Delta(\mathcal{P}_t,\mathcal{A}_t) &\leq 2\sum_{i\in\mathcal{P}_t} w_i|\hat{r}_i(t) - r_i| - M \\ &\leq 2\sum_{i\in\mathcal{P}_t}(w_i|\hat{r}_i(t) - r_i| - \frac{M}{2|\mathcal{P}_t|}) \\ &= 2\sum_{i\in\mathcal{P}_t}\Big(w_i|\hat{r}_i(t) - r_i| - \frac{M}{2L}\Big),\end{aligned}$$

Then, using (23), the expression above becomes:

$$\Delta(\mathcal{P}_t, \mathcal{A}_t) \le 2 \sum_{i \in \mathcal{P}_t} \Big(w_i \cdot \min\{2\sqrt{\frac{3 \ln t}{2 n_i(t-1)}}, 1\} - \frac{M}{2L} \Big)$$
$$= \sum_{i \in \mathcal{P}_t} \Big(w_i \cdot \min\{\sqrt{\frac{24 \ln t}{n_i(t-1)}}, 2\} - \frac{M}{L} \Big).$$

Finally, we note that when $n_i(t-1) \le l_T(M)$, then $w_i \cdot \min\{\sqrt{\frac{24 \ln t}{n_i(t-1)}}, 2\} - \frac{M}{L} \le \kappa_T(M, n_i(t-1))$. However, when $n_i(t-1) > l_T(M)$ then $w_i \cdot \min\{\sqrt{\frac{24 \ln t}{n_i(t-1)}}, 2\} \le \sqrt{\frac{24 \ln t}{n_i(t-1)}} \le \frac{M}{L}$, thus $\min\{\sqrt{\frac{24 \ln t}{n_i(t-1)}}, 2\} - \frac{M}{L} \le 0 = \kappa_T(M, n_i(t-1))$, which completes the proof. □

C. Proof of Theorem 2

We next present the complete proof of the constraint violation bound.

Theorem 2. Under the same conditions as **Theorem 1**, the total long-term violation $V(T)$ defined in (9) satisfies the following sublinear bound:

$$V(T) \le 2\sqrt{24KLT \ln T} + 2\pi^2 KL + 2L\sqrt{T}.$$

Proof. We first decompose the total long-term violation into two parts:

$$V(T) = V_{cost}(T) + V_{fid}(T) \tag{30}$$

We define the long-term cost violation and long-term fidelity violation as follows:

$$V_{cost}(T) = \mathbb{E}\left[\left(\sum_{t=1}^{T} \left(\sum_{i \in \mathcal{P}_t} \mu_i(t) - B \right) \right)^+ \right], \tag{31}$$

$$V_{fid}(T) = \mathbb{E}\left[\left(\sum_{t=1}^{T} \sum_{i \in \mathcal{P}_t} [F_{\min} - F_i(t)] \right)^+ \right]. \tag{32}$$

At each time slot t, the oracle guarantees that the estimated cost and fidelity satisfy:

$$\begin{aligned} &\sum_{i \in \mathcal{P}_t} \check{\mu}_i(t) \le B, \\ &\hat{F}_i(t) \ge F_{\min}, \forall i \in \mathcal{P}_t. \end{aligned} \tag{33}$$

Using (33), we decompose the violations into two parts.

$$\sum_{i \in \mathcal{P}_t} \mu_i(t) - B = \sum_{i \in \mathcal{P}_t} (\mu_i(t) - \check{\mu}_i(t)) + \underbrace{\sum_{i \in \mathcal{P}_t} \check{\mu}_i(t) - B}_{\le 0},$$

$$\sum_{i \in \mathcal{P}_t} [F_{\min} - F_i(t)] = \sum_{i \in \mathcal{P}_t} [(\hat{F}_i(t) - F_i(t)) + \underbrace{(F_{\min} - \hat{F}_i(t))}_{\le 0}], \tag{34}$$

Therefore, we have:

$$\sum_{i \in \mathcal{P}_t} \mu_i(t) - B \le \sum_{i \in \mathcal{P}_t} (\mu_i(t) - \check{\mu}_i(t)), \tag{35}$$

$$\sum_{i \in \mathcal{P}_t} [F_{\min} - F_i(t)] \le \sum_{i \in \mathcal{P}_t} (\hat{F}_i(t) - F_i(t)). \tag{36}$$

Recall that $\check{\mu}_i(t) = \bar{\mu}_i(t-1) - \mathrm{rad}_i(t)$ and $\hat{F}_i(t) = \bar{F}_i(t-1) + \mathrm{rad}_i(t)$. Thus the cumulative deviation is:

$$\begin{aligned} \sum_{t=1}^{T} \sum_{i \in \mathcal{P}_t} (\mu_i(t) - \check{\mu}_i(t)) &= \underbrace{\sum_{t=1}^{T} \sum_{i \in \mathcal{P}_t} (\mu_i(t) - \mu_i)}_{S_T^{cost}} \\ &+ \underbrace{\sum_{t=1}^{T} \sum_{i \in \mathcal{P}_t} (\mu_i - \bar{\mu}_i(t-1))}_{\mathcal{E}_T^{cost}} + \underbrace{\sum_{t=1}^{T} \sum_{i \in \mathcal{P}_t} \mathrm{rad}_i(t)}_{\mathcal{R}_T}. \end{aligned} \tag{37}$$

and

$$\begin{aligned} \sum_{t=1}^{T} \sum_{i \in \mathcal{P}_t} (\hat{F}_i(t) - F_i(t)) &= \underbrace{\sum_{t=1}^{T} \sum_{i \in \mathcal{P}_t} (F_i - F_i(t))}_{S_T^{fid}} \\ &+ \underbrace{\sum_{t=1}^{T} \sum_{i \in \mathcal{P}_t} (\bar{F}_i(t-1)) - F_i)}_{\mathcal{E}_T^{fid}} + \underbrace{\sum_{t=1}^{T} \sum_{i \in \mathcal{P}_t} \mathrm{rad}_i(t)}_{\mathcal{R}_T}. \end{aligned} \tag{38}$$

Hence:

$$\begin{aligned} V_{cost}(T) &\le \mathbb{E}[(S_T^{cost})^+] + \mathbb{E}[(\mathcal{E}_T^{cost})^+] + \mathbb{E}[(\mathcal{R}_T)^+], \\ V_{fid}(T) &\le \mathbb{E}[(S_T^{fid})^+] + \mathbb{E}[(\mathcal{E}_T^{fid})^+] + \mathbb{E}[(\mathcal{R}_T)^+]. \end{aligned} \tag{39}$$

To bound the stochastic fluctuation term $\mathbb{E}[(S_T^{cost})^+]$ and $\mathbb{E}[(S_T^{fid})^+]$, we apply Lemma 6 to $W_t^{cost} = \sum_{i \in \mathcal{P}_t} (\mu_i(t) - \mu_i)$ and $W_t^{fid} = \sum_{i \in \mathcal{P}_t} (F_i - F_i(t))$. We verify that $\{W_t^{cost}\}$ and $\{W_t^{fid}\}$ are martingale difference sequence:

$$\begin{aligned} \mathbb{E}[W_t^{cost} | H_{t-1}] &= \mathbb{E}\left[\sum_{i \in \mathcal{P}_t} (\mu_i(t) - \mu_i) \middle| H_{t-1} \right] \\ &= \sum_{i \in \mathcal{P}_t} \mathbb{E}[\mu_i(t) - \mu_i | H_{t-1}] = \sum_{i \in \mathcal{P}_t} (\mathbb{E}[\mu_i(t)] - \mu_i) = 0. \end{aligned}$$

$$\begin{aligned} \mathbb{E}[W_t^{fid} | H_{t-1}] &= \mathbb{E}\left[\sum_{i \in \mathcal{P}_t} (F_i - F_i(t)) \middle| H_{t-1} \right] \\ &= \sum_{i \in \mathcal{P}_t} \mathbb{E}[F_i - F_i(t) | H_{t-1}] = \sum_{i \in \mathcal{P}_t} (F_i - \mathbb{E}[F_i(t)]) = 0. \end{aligned}$$

and

$$|W_t^{cost}| = \left| \sum_{i \in \mathcal{P}_t} (\mu_i(t) - \mu_i) \right| \le \sum_{i \in \mathcal{P}_t} |\mu_i(t) - \mu_i| \le L.$$

$$|W_t^{fid}| = \left| \sum_{i \in \mathcal{P}_t} (F_i - F_i(t)) \right| \le \sum_{i \in \mathcal{P}_t} |F_i - F_i(t)| \le L.$$

Therefore, we have:

$$\mathbb{E}[(S_T^{cost})^+] \le L\sqrt{T}. \tag{40}$$

$$\mathbb{E}[(S_T^{fid})^+] \le L\sqrt{T}. \tag{41}$$

To bound the estimation error term $\mathbb{E}[(\mathcal{E}_T^{cost})^+]$ and $\mathbb{E}[(\mathcal{E}_T^{fid})^+]$, we decompose them into the $\mathcal{N}_t$ event and their complement:

$$\begin{aligned}\mathbb{E}[(\mathcal{E}_T^{cost})^+] \leq & \mathbb{E}\left[\sum_{t=1}^{T}\mathbb{I}(\mathcal{N}_t)\sum_{i\in\mathcal{P}_t}\mathrm{rad}_i(t)\right] \\ & + \mathbb{E}\left[\sum_{t=1}^{T}\mathbb{I}(\neg\mathcal{N}_t)\cdot L\right]\end{aligned}$$

and

$$\begin{aligned}\mathbb{E}[(\mathcal{E}_T^{fid})^+] \leq & \mathbb{E}\left[\sum_{t=1}^{T}\mathbb{I}(\mathcal{N}_t)\sum_{i\in\mathcal{P}_t}\mathrm{rad}_i(t)\right] \\ & + \mathbb{E}\left[\sum_{t=1}^{T}\mathbb{I}(\neg\mathcal{N}_t)\cdot L\right]\end{aligned}$$

Therefore, on the event $\mathcal{N}_t$, we have:

$$\begin{aligned}&\mathbb{E}\left[\sum_{t=1}^{T}\mathbb{I}(\mathcal{N}_t)\sum_{i\in\mathcal{P}_t}\mathrm{rad}_i(t)\right] \leq \mathbb{E}\left[\sum_{t=1}^{T}\sum_{i\in\mathcal{P}_t}\mathrm{rad}_i(t)\right] \\ &= \mathbb{E}\left[\sum_{t=1}^{T}\sum_{i\in\mathcal{P}_t}\sqrt{\frac{3\ln t}{2n_i(t-1)}}\right] \\ &\leq \mathbb{E}\left[\sum_{i\in\mathcal{P}_t}\sum_{s=1}^{n_i(T)-1}\sqrt{\frac{3\ln T}{2s}}\right] \leq \mathbb{E}\left[\sum_{i\in\mathcal{P}_t}\sqrt{6n_i(T)\ln T}\right] \\ &\leq \mathbb{E}\left[\sum_{i=1}^{K}\sqrt{6n_i(T)\ln T}\right] \leq \mathbb{E}\left[\sqrt{6K\ln T}\cdot\sqrt{\sum_{i=1}^{K}n_i(T)}\right] \\ &= \sqrt{6KLT\ln T}\end{aligned} \tag{42}$$

On the complement $\neg\mathcal{N}_t$, we have:

$$\begin{aligned}\mathbb{E}\left[\sum_{t=1}^{T}\mathbb{I}(\neg\mathcal{N}_t)\cdot L\right] &\leq L\cdot\sum_{t=1}^{T}\mathbb{P}(\neg\mathcal{N}_t) \leq L\sum_{t=1}^{T}\frac{6K}{t^3} \\ &\leq L\sum_{t=1}^{T}\frac{6K}{t^2} \leq \pi^2 KL.\end{aligned} \tag{43}$$

Combing (42) and (43), we have:

$$\mathbb{E}[(\mathcal{E}_T^{cost})^+] \leq \sqrt{6KLT\ln T} + \pi^2 KL. \tag{44}$$

$$\mathbb{E}[(\mathcal{E}_T^{fid})^+] \leq \sqrt{6KLT\ln T} + \pi^2 KL. \tag{45}$$

To bound the bonus term $\mathbb{E}[(\mathcal{R}_T)^+]$, we use the conclusion in (42):

$$\mathbb{E}[(\mathcal{R}_T)^+] = \mathbb{E}\left[\sum_{t=1}^{T}\sum_{i\in\mathcal{P}_t}\mathrm{rad}_i(t)\right] \leq \sqrt{6KLT\ln T} \tag{46}$$

Therefore, the long-term cost and fidelity violation satisfy:

$$V_{cost}(T) \leq \sqrt{24KLT\ln T} + \pi^2 KL + L\sqrt{T},$$

and

$$V_{fid}(T) \leq \sqrt{24KLT\ln T} + \pi^2 KL + L\sqrt{T}.$$

Combining the established bounds for the cost violation and the fidelity violation, we immediately obtain:

$$\begin{aligned}V(T) &= V_{cost}(T) + V_{fid}(T) \\ &\leq 2\sqrt{24KLT\ln T} + 2\pi^2 KL + 2L\sqrt{T}.\end{aligned}$$

□

Lemma 6. Let $\{H_t\}_{t=0}^T$ be a filtration, and $\{W_t\}_{t=1}^T$ a martingale difference sequence adapted to $\{H_t\}$, satisfying $\mathbb{E}[W_t \mid H_{t-1}] = 0$ for every time slot t. Suppose there exists a constant $C > 0$ such that $|W_t| \leq C$ almost surely for every time slot t. Define the partial sum as $S_T = \sum_{t=1}^T W_t$. Then we have:

$$\mathbb{E}[(S_T)^+] \leq C\sqrt{T}.$$

Proof. For any $\lambda > 0$, the standard Azuma-Hoeffding inequality for martingale differences with uniform bound C gives:

$$\mathbb{P}(S_T \geq \lambda) \leq \exp\Big(-\frac{\lambda^2}{2TC^2}\Big).$$

For any real-valued random variable X, the expectation of its positive part can be rewritten as an integral over tail probabilities: $\mathbb{E}[X^+] = \int_0^\infty \mathbb{P}(X \geq \lambda)d\lambda$. Set $X = S_T$ and plug the exponential upper bound, we have:

$$\mathbb{E}[(S_T)^+] \leq \int_0^\infty \exp\Big(-\frac{\lambda^2}{2TC^2}\Big)d\lambda = \frac{1}{2}\sqrt{2\pi TC^2} \leq C\sqrt{T}$$

□